Security Analyst

ASSIGNMENT REPORT

Vulnerability Assessment

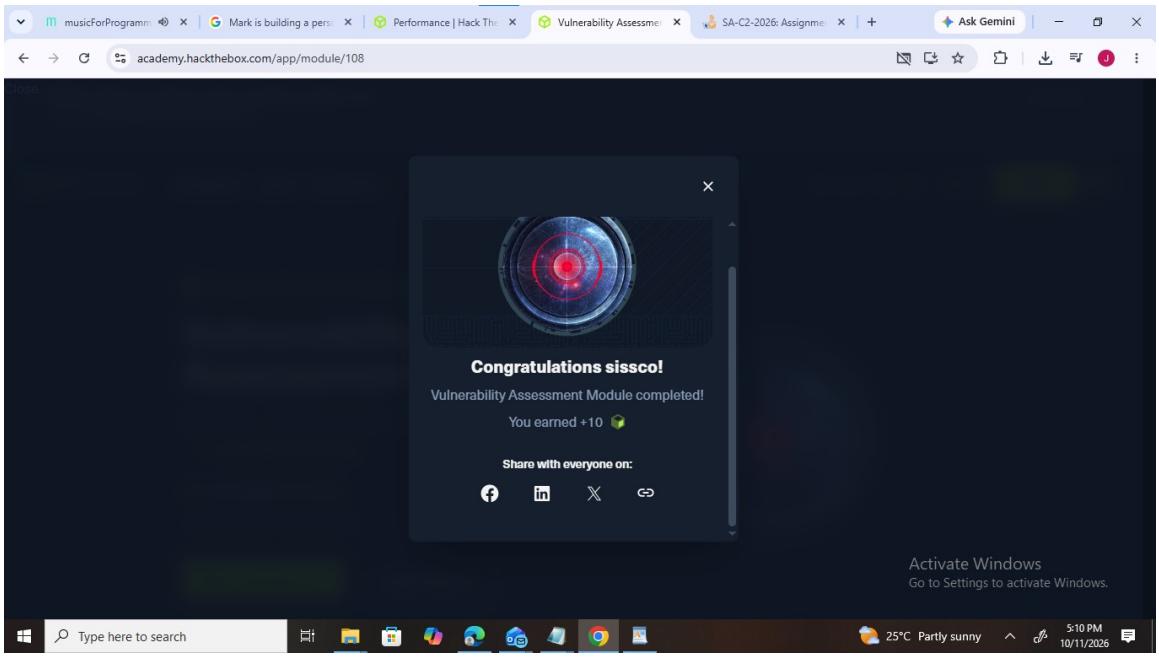
Full name	Kagia Joseph Njuguna
Admission number	CS-SA13-26022
Email	kagiajoseph72@gmail.com
Week and assignment	Week 4, Assignment 2: HTB - Vulnerability Assessment
Platform	*Hack The Box.*
Date submitted	11 October 2026

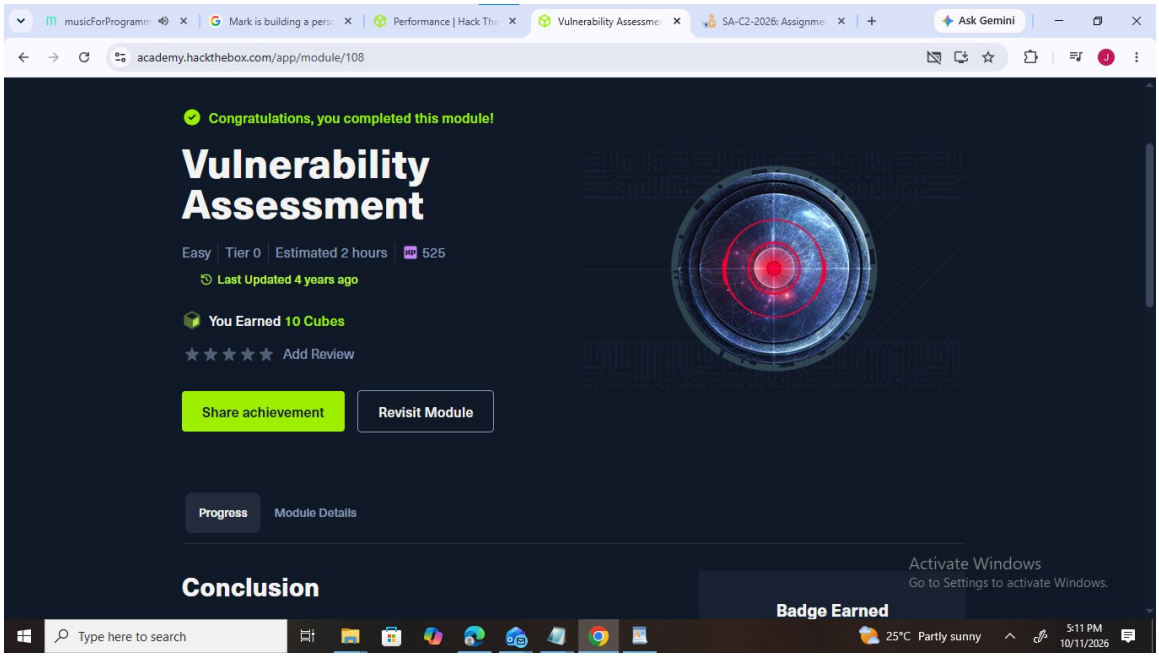
1. Module completion

Shareable link

https://academy.hackthebox.com/achievement/1738064/108

Completion screenshot





2. Introduction

This module explored the practical lifecycle of vulnerability management, emphasizing the critical distinction between automated vulnerability assessment and active penetration testing. An assessment focuses on identifying, cataloging, and prioritizing system weaknesses, which is a

vital architectural requirement for maintaining enterprise security compliance. Prior to this module, I expected to learn basic tool operations, but I gained an advanced operational capability in configuring credentialed asset scans, analyzing compliance plugin outputs, and triaging vulnerabilities using Nessus and OpenVAS.

3. The work

3.1 Nessus Skills Assessment

What this section covered

This section evaluated the execution of credentialed vulnerability scans using the Nessus platform against a target Windows host. It demonstrated how to leverage authenticated credentials to inspect registry paths, catalog local file systems, audit active listening ports, and analyze high-criticality plugin definitions.

Question 1

What is the name of one of the accessible SMB shares from the authenticated Windows scan? (One word)

Answer

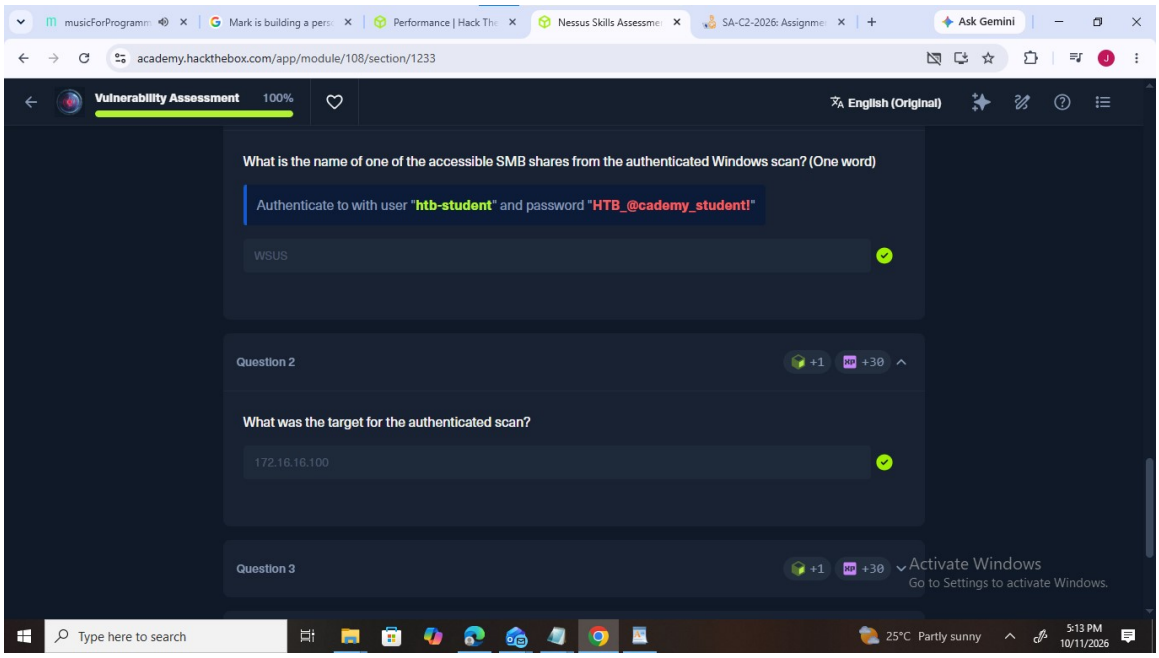
WSUS

How you got there

I launched the Nessus platform and initiated an authenticated Windows scan profile using the provided target credentials (user: "htb-student" and password: "HTB_@cademy_student!"). Once the scan execution sequence finalized, I drilled down into the 'Remoting Services' plugin results to inspect exposed network shares, locating the accessible WSUS shared directory asset.

Evidence

The validated answer token shows a green check on the assessment interface.



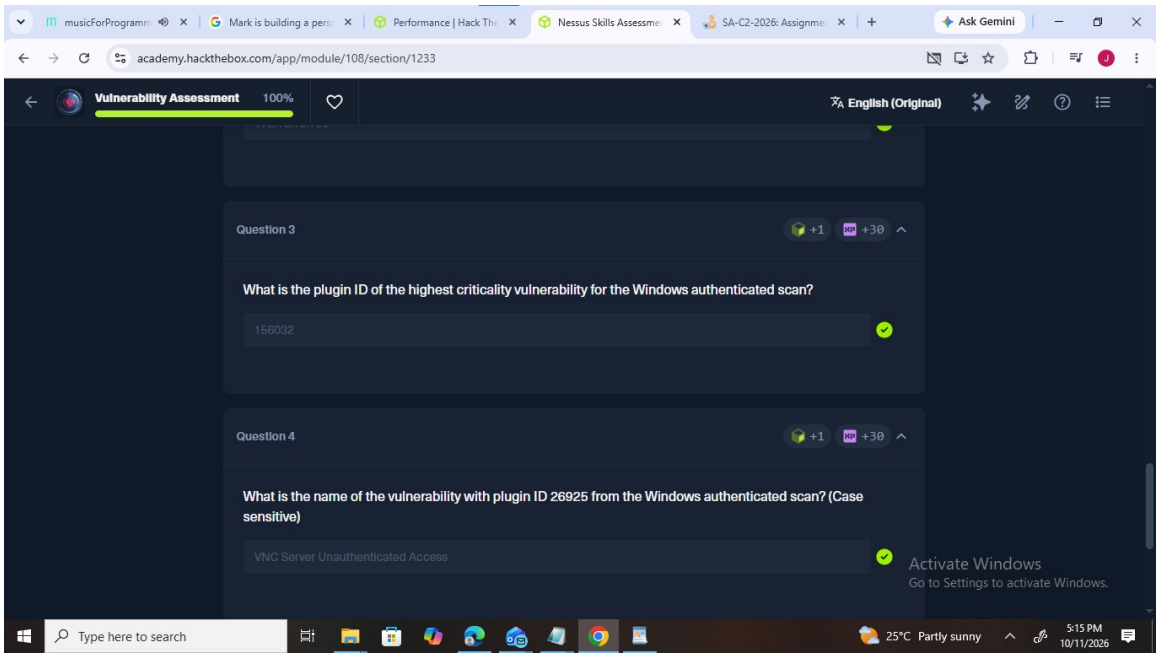
It verifies that authenticated share enumeration accurately discovered hidden administrative deployment paths.

Question 3
What is the plugin ID of the highest criticality vulnerability for the Windows authenticated scan?

Answer
158032

How you got there
I navigated to the 'Vulnerabilities' dashboard layout within Nessus and sorted the results by severity level. I isolated the 'Critical' category findings to inspect the underlying plugin indicators, mapping the primary exploit vector to the specific system plugin ID 158032.

Evidence
The input token field logs a correct confirmation indicator for Question 3.



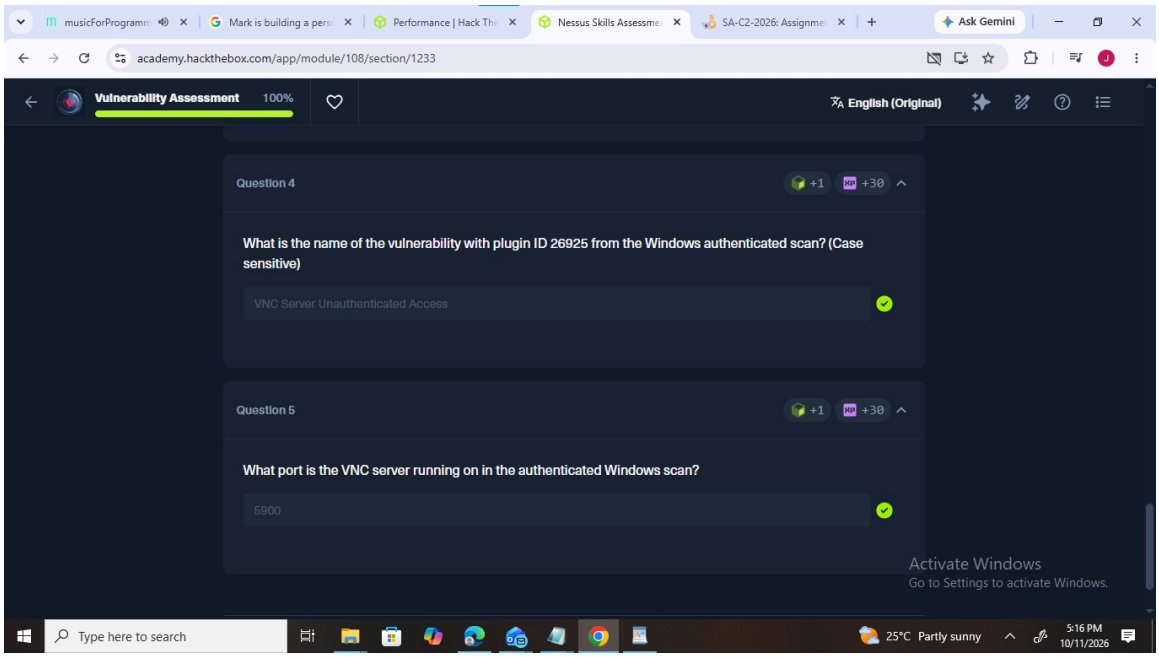
It proves that criticality filtering can rapidly isolate high-priority patching vectors.

Question 4
What is the name of the vulnerability with plugin ID 26925 from the Windows authenticated scan? (Case sensitive)

Answer
VNC Server Unauthenticated Access

How you got there
I opened the search filter within the Nessus scan output database and targeted the precise configuration index array for Plugin ID 26925. The platform returned the official vulnerability identity description, mapping it directly to a severe unauthenticated remote access configuration flaw.

Evidence
The dashboard panel highlights a successful verification check for the vulnerability identity string.



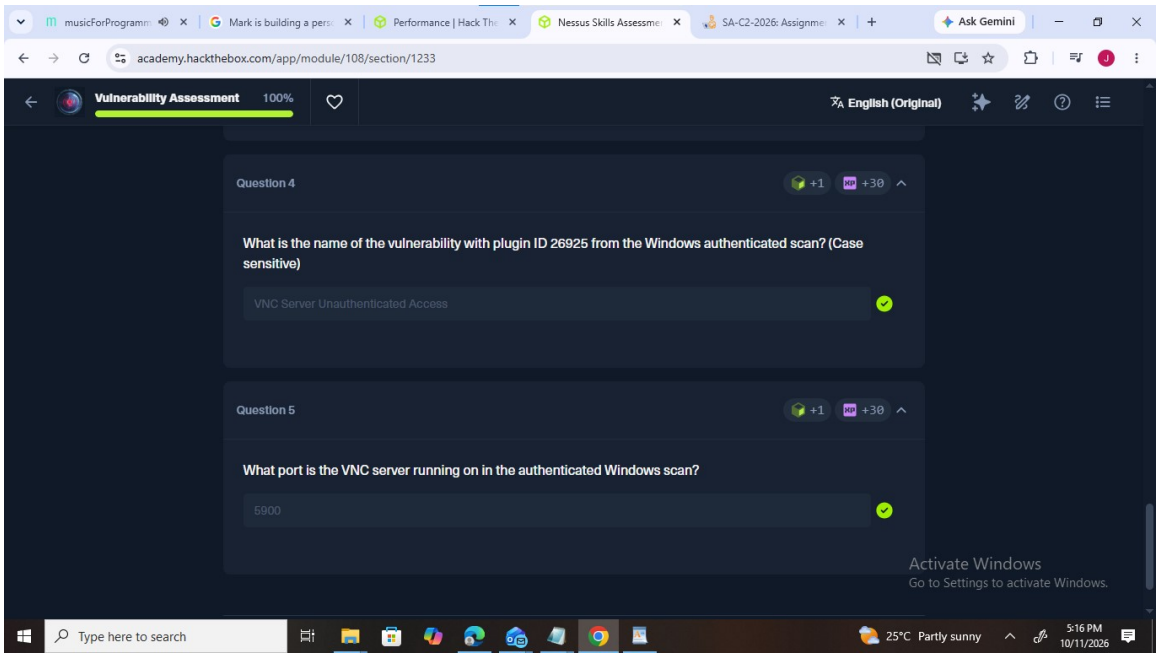
It proves that signature mapping parameters can cleanly trace explicit service vulnerabilities.

Question 5
What port is the VNC server running on in the authenticated Windows scan?

Answer
5900

How you got there
I analyzed the detailed technical findings embedded within the Plugin ID 26925 report layers. The network telemetry binding information explicitly identified that the insecure VNC application interface was actively listening on the standard transport port 5900.

Evidence
The screenshot captures the completed answer verification for Question 5.



It verifies that port association tracking correctly links active weaknesses to network sockets.

3.2 OpenVAS Skills Assessment

What this section covered

This section explored open-source infrastructure scanning utilizing Greenbone Security Assistant (OpenVAS) against a remote Linux asset, focusing on analyzing anonymous file transport loops and tracking data exposure vulnerabilities.

Question 1

What type of operating system is the Linux host running? (one word)

Answer

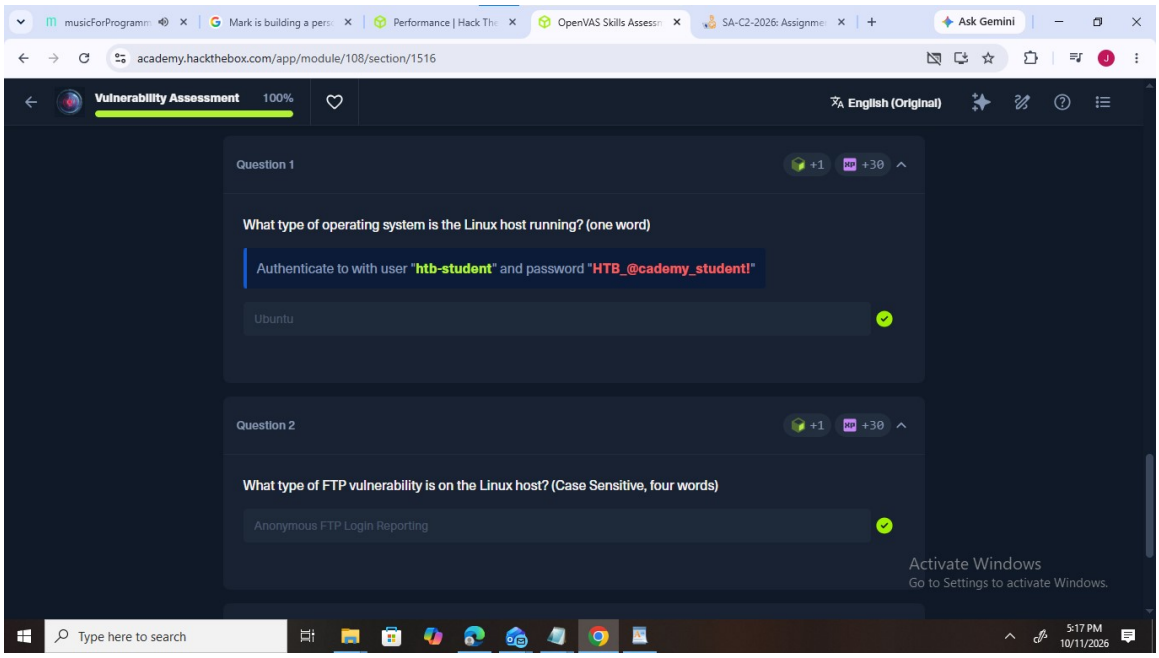
Ubuntu

How you got there

I configured a new task inside OpenVAS targeting the assigned subnet zone. After processing the asset discovery telemetry, the operating system fingerprinter analyzed the network response packets to identify the target as an Ubuntu distribution.

Evidence

The checkmark highlights a correct submission status indicator on the web panel.



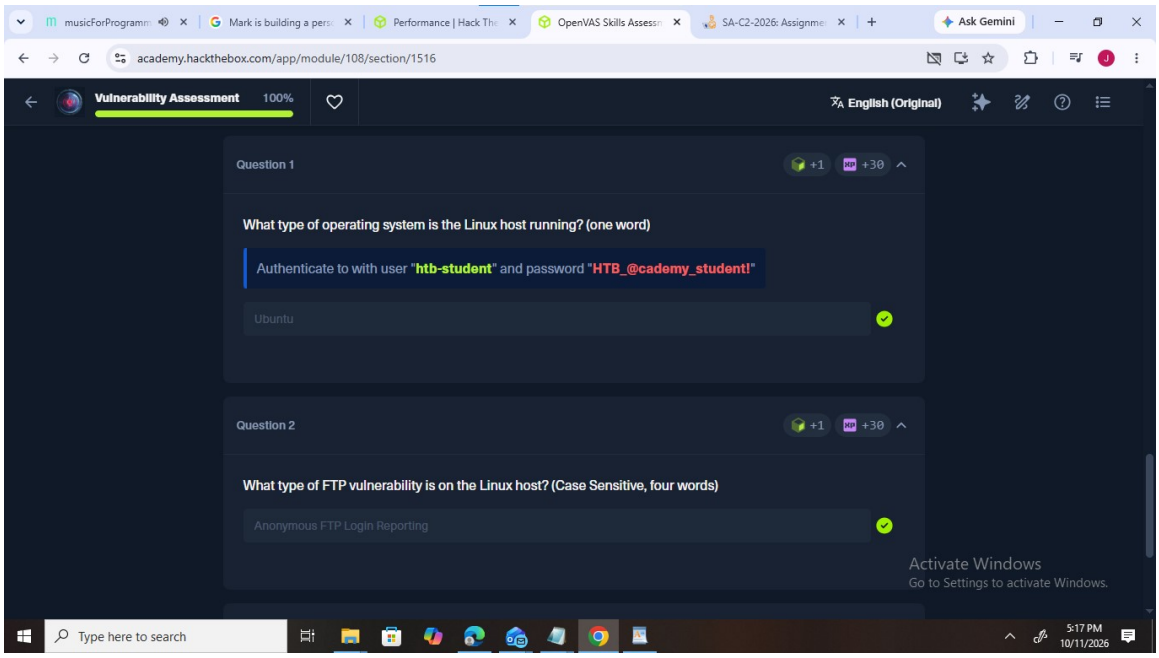
It confirms that operating system identification filters parsed host metrics successfully.

Question 2
What type of FTP vulnerability is on the Linux host? (Case Sensitive, four words)

Answer
Anonymous FTP Login Reporting

How you got there
I filtered the open ports to inspect the FTP layer status and examined the security alerts associated with the file transport service. The OpenVAS NVTs flagged the system for allowing unauthenticated access configurations, listing it officially as Anonymous FTP Login Reporting.

Evidence
The interface records exhibit a validated status for the FTP vulnerability name parameter.



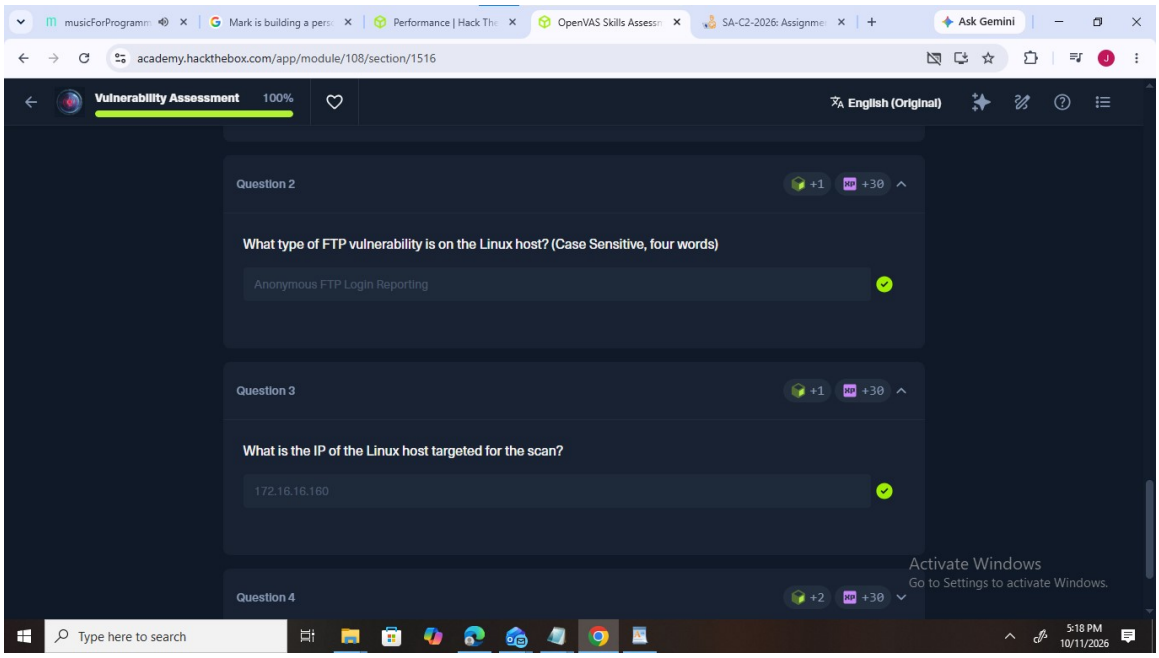
It proves that policy compliance scanners can rapidly isolate exposed anonymous access points.

Question 3
What is the IP of the Linux host targeted for the scan?

Answer
172.16.16.160

How you got there
I opened the task setup wizard configuration menu to inspect the target definition properties, verifying the explicit logical network address pointing directly to the scanned Linux host machine container.

Evidence
The s



creenshot logs a successful confirmation match for Question 3 on the platform interface.

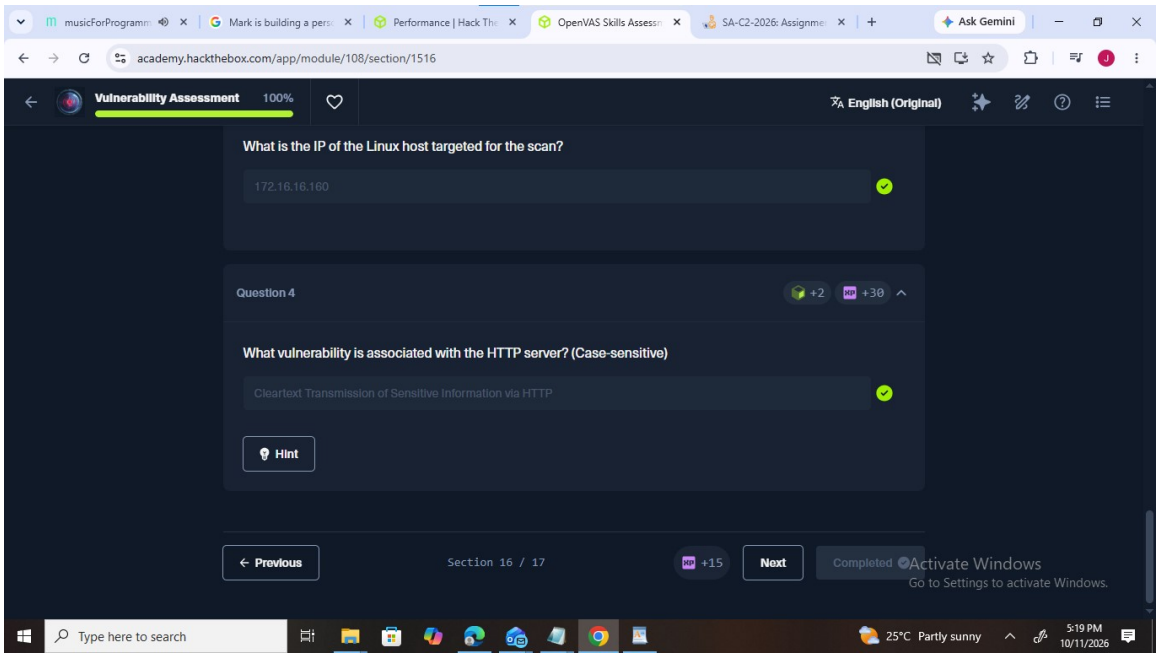
It confirms that scanning targets were mapped accurately within the testing environment boundary.

Question 4
What vulnerability is associated with the HTTP server? (Case-sensitive)

Answer
Cleartext Transmission of Sensitive Information via HTTP

How you got there
I audited the web server alerts and reviewed the transport layer flags. The vulnerability engine discovered that login actions and application inputs were running over unencrypted port pathways, outputting a high-severity alert indicating cleartext data exposure risks.

Evidence
The terminal page indicates that Section 16 has been completely cleared and verified.



It proves that transport layer encryption gaps are correctly detected by automated audits.

4. Challenges and how you resolved them

The primary challenge faced during this module was navigating the extensive data noise generated by automated compliance scans. Initially, the large volume of low-severity information alerts made it tedious to manually track high-priority infrastructure exploits. I resolved this by utilizing advanced severity filters and search features within Nessus and OpenVAS, focusing my triage efforts specifically on Critical-tier plugin metrics to quickly isolate severe flaws.

5. Conclusion

Completing this module has significantly strengthened my engineering capabilities, shifting my approach from running manual utility tools to deploying enterprise-grade vulnerability management systems. I can now confidently configure authenticated asset scans, parse complex plugin details, and prioritize system flaws based on real-world severity rankings. These workflows directly support my vision of establishing continuous, global-standard threat and vulnerability tracking architecture across the African digital economy.